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Original Article

High serum concentrations of the acyclovir main metabolite 9-carboxymethoxymethylguanine in renal failure patients with acyclovir-related neuropsychiatric side effects: an observational study

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Abstract

Background. Acyclovir (ACV) has been used for over two decades to treat herpes virus infections. Serious neurological adverse side effects have occurred during ACV treatment in patients with renal failure, but the cause of the symptoms remains unknown. We hypothesized that increased concentrations of the ACV main metabolite 9-carboxymethoxymethylguanine (CMMG) correlated to these symptoms.

Methods. We conducted an observational study from 1991 to mid 1999 based on samples sent for analysis of ACV concentration from various hospital departments in Sweden. Patients with neuropsychiatric symptoms (NS+, $n=49$) were compared with patients without symptoms (NS−, $n=44$). ACV and CMMG concentrations were analysed by HPLC. Medical records were analysed for symptoms and compared with pertinent cases identified from Medline.

Results. The serum CMMG levels were significantly higher in the NS+ group (mean = 34.1 μmol/l, 95% confidence interval 23.4–46.1) compared with the NS− group (mean = 4.7 μmol/l, 95% confidence interval 3.3–6.6; $P<0.001$). CMMG was the strongest predictor in a receiver-operating characteristics curve analysis (ROC), based on 77 patients, of ACV-related neuropsychiatric symptoms. The ROC curve for CMMG demonstrated that neuropsychiatric symptoms could be predicted with a sensitivity of 91% and a specificity of 93% with the use of a cut-off value of 10.8 μmol/l of CMMG. Thirty-five of 49 patients in the NS+ group showed levels exceeding this concentration compared with only three of 44 of patients in the NS− group ($P<0.001$). ACV exposure, ACV concentration, creatinine clearance and creatinine concentration were weaker but statistically significant predictors. Haemodialysis reduced CMMG and ACV levels and relieved the symptoms.

Conclusions. The determination of CMMG levels in serum may be a useful tool in supporting the diagnosis of ACV-associated neuropsychiatric symptoms. Furthermore, the monitoring of CMMG levels may prevent the emergence of symptoms.

Keywords: acyclovir; adverse effects; 9-carboxymethoxymethylguanine; neuropsychiatric; renal failure

Introduction

Neuropsychiatric symptoms appearing during acyclovir (ACV) therapy have repeatedly been recognized and are reported in patients with acute or chronic renal failure [1,2]. Such symptoms have also been reported in ACV-treated patients with various malignancies and in bone marrow transplant recipients [3,4]. The symptoms have been observed after i.v. [3] as well as oral [5] treatment with ACV. Recently, valaciclovir (VACV) has also been associated with similar side effects [6]. The mechanism behind the symptoms is not known. Clinically, central nervous system (CNS) symptoms due to ACV treatment have been misinterpreted as symptoms of herpes encephalitis and contributed to the physician's decision to increase the ACV dose, rather than decrease it [7]. It has not been possible to establish a clear relation between serum concentrations of ACV and neuropsychiatric symptoms. Some authors have correlated high concentrations of ACV to neuropsychiatric events [8], while according to other

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case reports, patients have shown CNS symptoms despite normal or low concentrations of ACV [9]. The possible influence of ACV metabolites on CNS symptoms has been discussed previously [10] but not *de facto* investigated.

ACV is probably metabolized by alcohol dehydrogenase and aldehyde dehydrogenase to 9-carboxymethoxymethylguanine (CMMG) and to a smaller extent to 8-hydroxy-9-(2-hydroxyethoxymethyl) guanine (8-OH-ACV) [11]. In patients without renal failure, intravenously administered ACV is excreted in the urine 62–91% unchanged and 8–14% as CMMG. A larger proportion of the ACV dose is excreted as CMMG in the urine when renal function is impaired [12]. The mean half-life of ACV is increased from 3.8 h in normal subjects to 19.5 h in patients with chronic renal failure [13]. Thus, the steady-state concentration of ACV is expected to increase as renal function declines. CMMG in patients with decreased renal function has not been evaluated.

The aim of this study was to determine if there was a correlation between increased serum concentrations of CMMG and neuropsychiatric symptoms.

Subjects and methods

Subjects

The study was based on blood samples from 93 patients sent to the Department of Clinical Pharmacology, Huddinge University Hospital, Stockholm, Sweden for analysis of serum concentrations of ACV from November 1991 until June 1999. The samples were collected at Departments of Infectious Diseases, Haematology and Nephrology in Swedish hospitals. We reviewed the patient charts to acquire information on age, sex, serum creatinine, estimated creatinine clearance, weight, dose, administration of ACV, ACV exposure, the lag time from previous dose until sample and on neuropsychiatric symptoms. The symptoms were classified as present (NS+) or absent (NS−) by a neurologist who had no knowledge of the patients' ACV and CMMG concentrations. Patients were included in the NS+ group even if other causes could be related to the symptoms such as meningitis or encephalitis. Next, the NS+ group was subdivided into two subgroups. The first subgroup included patients in whom no other cause of the CNS symptoms than ACV was found. The second subgroup included three different types of patients: (i) patients in whom CNS-infection was present, (ii) patients treated with drugs with a known risk of CNS-side effect or (iii) patients whose charts did not include information about the presence of symptoms when the blood sample was taken.

The NS+ group was also subject to further analysis of several aspects of the patient records including: the effect of dialysis treatment, the indication for ACV treatment, the lag time from start of treatment until appearance of symptoms, the pattern of symptoms and the outcome of the symptom episodes. We also analysed the results from the following clinical investigations: EEG, culture and PCR of cerebrospinal fluid, cranial computer tomography scan (CT) and magnetic resonance imaging (MRI). The research ethics committee of Huddinge University Hospital approved the study.

Assessment of adverse drug reactions

The neuropsychiatric symptoms compiled from the patient files were compared with published case reports and articles quoted in this article. The neuropsychiatric symptoms were categorized into six main groups reflecting specific functions of the nervous system. Other symptoms, such as disturbances in lung function and acute renal failure, were also collated. Measures of metabolic disturbances were not systematically collated, but were often part of the clinical investigations.

Samples

The first serum concentration obtained from each patient was used in the concentration–symptom analysis. The first sample was selected for two reasons; many patients had only one sample drawn, secondly, subsequent concentrations may have been influenced by the result from the first sample due to e.g. changed dosing. Pre- and post-dialysis ACV and CMMG concentrations were obtained from nine patients. The serum concentrations were determined by solid phase extraction followed by high-performance liquid chromatography with fluorescence detection [14]. The limit of detection was 0.12 μmol/l.

ACV exposure

To estimate ACV exposure in each patient an oral bioavailability (F) of 20% for 200-mg tablets, 13% for 400-mg tablets and 9% for 800-mg tablets, was assumed, based on earlier studies [15]. VACV was assumed to be 54% bioavailable [16]. Exposure was calculated using the equation for steady-state concentration (Css):

$$C_{SS} = \frac{F \times D}{CL \times \Delta t} = \frac{F \times D}{ACVCLn \times (0.0085 * CrCL + 0.15) \times \Delta t}$$

Here, F is bioavailability, D is dose (in μmol), ACVCLn is the ACV clearance of an individual with normal renal function, CrCL is estimated creatinine clearance (in ml/min) and Δt is the dose interval (in minutes). In this equation it is assumed that ACV clearance (CL) is proportional to the sum of creatinine clearance (CrCL) and a non-renal clearance component. The results are presented as a percentage of the exposure in a patient with CrCL = 100 ml/min who is treated with ACV 800 mg five times daily. In this way the exact value of ACVCLn need not be specified as it is eliminated in this calculation of relative exposure.

Statistics

Data on the concentrations of CMMG and ACV and the ACV exposure were analysed by Student's t-test. Doses were compared by Mann–Whitney U-test. Descriptive statistics are given as means ± SD and in the abstract as means with 95% confidence intervals. The frequency data was analysed using χ^2 test. The computer program Statistica version 6.0 (Statsoft, New York, NY) was used in the study. Conventional receiver-operating characteristic (ROC) curves were used to determine the cut-off point of CMMG and ACV concentrations, ACV exposure, creatinine concentration and creatinine clearance with the highest sensitivity and specificity with respect to distinguishing patients with neuropsychiatric symptoms from those without such symptoms. A calculated area under the curve of 1.0 indicates that the laboratory test is a perfect indicator of neuropsychiatric symptoms, and an area of 0.5 is equivalent to chance.

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Creatinine clearance

Each adult patient's creatinine clearance was calculated using the MDRD method [17], the children's using Schwartz's estimate [18].

Results

Classification of patients

Sufficient data for classification was obtained in 93 patients. There were 49 patients with neuropsychiatric symptoms (NS+) and 44 without (NS−). Demographic characteristics of these patients are given in Table 1. In 33 of the 49 cases of the NS+ group, there were no signs of CNS infections and no drug causing CNS side effects was given concomitantly. Among the remaining 16 patients with neuropsychiatric symptoms, seven showed clinical and/or laboratory signs of herpes encephalitis, three showed signs of unspecified viral encephalitis, one showed clinical and laboratory signs of bacterial meningitis and one was treated with drugs known to cause CNS side effects. For the remaining four patients the charts lacked information about when the neuropsychiatric symptoms appeared in relation to the time of the blood sample. Renal insufficiency (creatinine clearance <60 ml/min) was present in 46 of 49 patients in the NS+ group and in 36 of 44 in the NS− group.

CMMG and ACV concentration and ACV exposure

The mean CMMG concentration in the NS+ and NS− group was 34.1 ± 39.4 and 4.7 ± 4.7 μmol/l, respectively ($P < 0.001$) (Figure 1A). The mean ACV concentration for the NS+ and NS− patients was 21.0 ± 30.7 and 7.2 ± 6.7 μmol/l, respectively ($P = 0.004$) (Figure 1B). Excluding the 16 patients in the NS+ mixed subgroup, a ROC curve had a cut-off value of 10.8 μmol/l of CMMG as an indicator of neuropsychiatric symptoms with a sensitivity of 91% and a specificity of 93%. The positive and negative predicted values were 91 and 93%, respectively. The results from ROC curve analyses are shown in Figure 2 and are summarized in Table 2. CMMG is a significantly stronger predictor than ACV exposure, ACV concentration, creatinine clearance and creatinine. The latter four variables were statistically significant predictors not differing from one another with respect to the AUC of the ROC curves. The ACV doses administered are shown in Table 1. Patients with neuropsychiatric symptoms had a significantly higher exposure ($P < 0.001$) to ACV than patients without symptoms (Figure 3).

Table 1. Demographic characteristics of patients with and without neuropsychiatric symptoms

	Symptoms ($n=49$)	No symptoms ($n=44$)	Difference (P-values)
Age (years)	59 ± 15	52 ± 22	0.05
Female sex	24	18	0.63
Weight (kg)	67 ± 18[a]	66 ± 20[b]	0.8
Creatinine (μmol/l)	476 ± 261	409 ± 312	0.24
Estimated creatinine clearance (ml/min)	19.5 ± 21.1	29.6 ± 27.4	0.04[c]
0–10	26	16	
11–25	15	10	
26–50	6	7	
>50	5	11	
ACV oral preparation (mg)			
100–400	2	17	
401–800	5	5	
801–1200	3	2	
1201–2000	4	3	
2001–4000	6	3	
VACV oral preparation (mg)			
500–1500	7	7	
1501–2000	2	1	
2001–3000	1	1	
ACV i.v. treatment (mg/kg)			
2–10	10	2	
11–20	4	2	
>20	5	1	
Summary prescribed daily dose (mg)			
ACV oral	1780 ± 1290	870 ± 892	<0.001[c]
VACV oral	1450 ± 725	1333 ± 791	0.65
ACV i.v.	1294 ± 1095	1088 ± 892	0.73
Time from the commencement of the treatment until the collection of the first sample (days)	11 (range 1–172)	9 (range 1–74)	0.63
The mean lag time from the previous dose until the first sample (h)	29 (range 4–136)	16 (range 1–96)	0.06[c]

Values are mean (±SD). ACV, acyclovir; VACV, valaciclovir.
[a]$n=44$.
[b]$n=40$.
[c]Value statistical significant if $P<0.05$.

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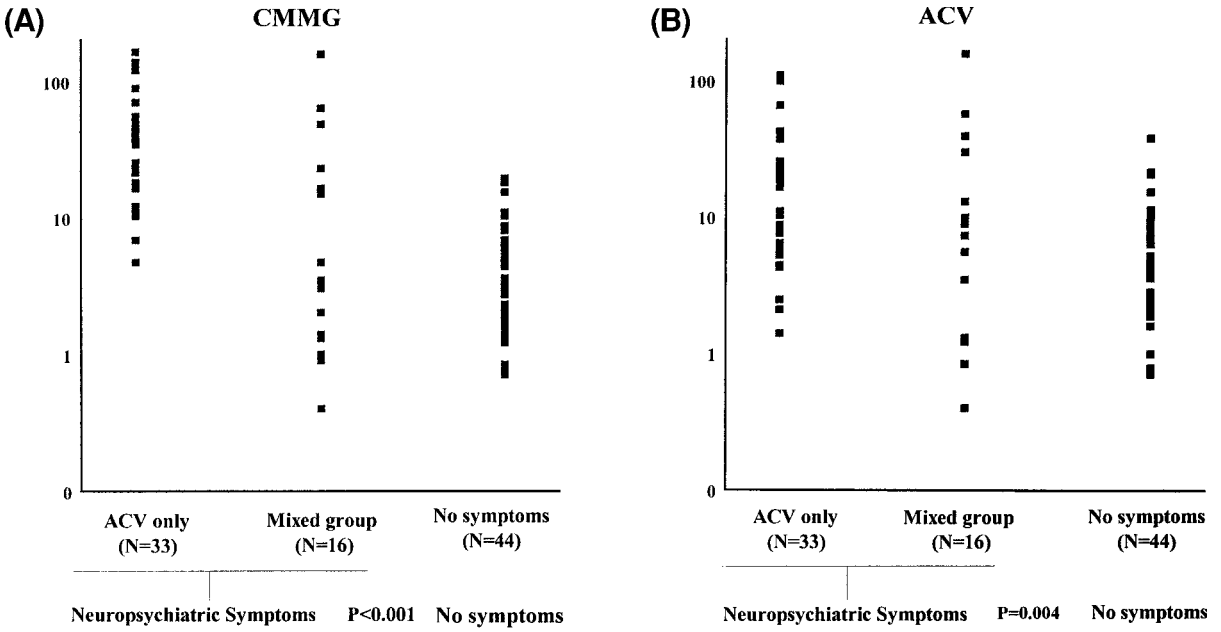


Fig. 1. Serum concentrations of (**A**) CMMG and (**B**) ACV in μmol/l in patients with and without neuropsychiatric symptoms. Six of the patients in the mixed subgroup had increased concentrations of CMMG and might have had symptoms both from their CNS infection and the ACV treatment. Haemodialysis treatment was not performed in these patients.

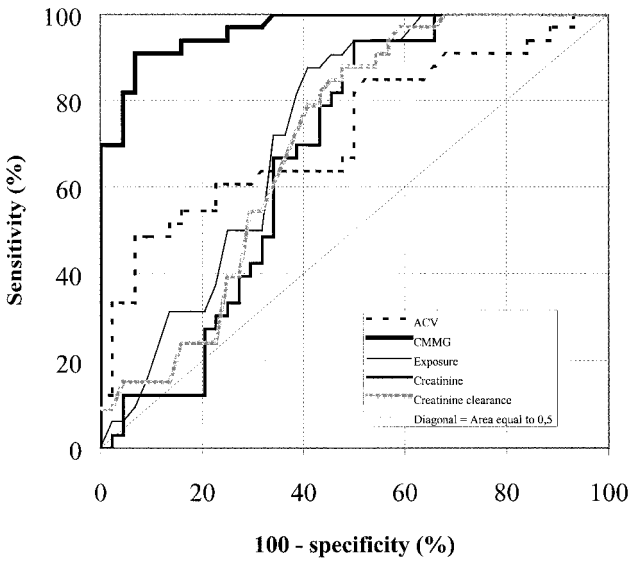


Fig. 2. ROC curves of CMMG and ACV levels, ACV exposure, creatinine clearance and creatinine. The curves show the fraction of true positive results (sensitivity) and false positive results (1– specificity) for various cut-off levels. See Table 2 for more information about the calculated areas.

Symptoms

A total of 74 different symptoms were described in our patient's charts and in previous reports. The symptoms found included, e.g. agitation, confusion, pronounced tiredness, lethargy, coma, dysarthria, myoclonus and visual and/or auditory hallucinations (Table 3). The time from the start of the treatment to the appearance of the symptoms was 1–2 days ($n = 35$).

Outcome

Thirty-nine patients recovered; 12/39 shortly after haemodialysis and 24/39 after withdrawal of ACV, dose reduction or treatment with i.v. fluid therapy when appropriate. Three patients with herpes encephalitis recovered after ACV treatment. Six patients died. Four of these patients died while unconscious and on ACV therapy. None of the six patients had signs of CNS infections. Three of them developed cerebral emboli, which was reported to be the cause of death. In addition, two surviving patients developed cerebral emboli and one patient with herpes encephalitis developed cerebral haemorrhage. One additional

Table 2. Results from ROC-curve analysis

Predictor	Cut-off value	Sensitivity (%)	Specificity (%)	AUC + (95% confidence interval)
CMMG	> 10.8 μmol/l	90.9	93.2	0.966 [0.897 0.993]
Exposure	> 62.67%	87.5	61.4	0.745 [0.632 0.838]
ACV	> 15.2 μmol/l	48.5	93.2	0.726 [0.613 0.822]
Creatinine clearance	⩽ 17 ml/min	87.9	52.3	0.692 [0.577 0.792]
Creatinine	> 251 μmol/l	93.9	50.0	0.679 [0.563 0.781]

AUC, area under the curve.

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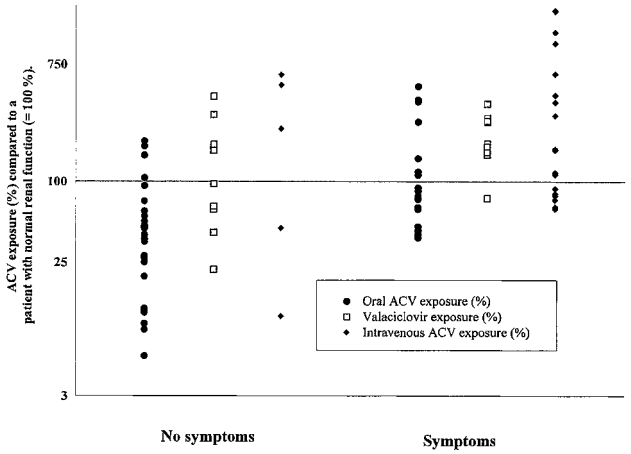


Fig. 3. ACV exposure based on preparation (oral, VACV or i.v. treatment), dose, bioavailability and renal function (see also the Subjects and methods section) in patients with and without neuropsychiatric symptoms. The exposure is expressed as a percentage of the exposure of a patient with normal renal function (CrCL = 100 ml/min) treated with ACV 800 mg five times daily.

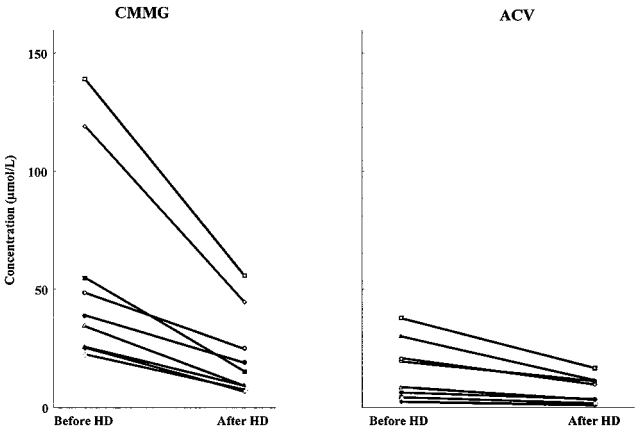


Fig. 4. Concentrations of CMMG and ACV serum concentrations in nine patients before and after a single 3–4 h haemodialysis (HD). Two patients had almost the same slope of the ACV curve. Each patient showed remarkable improvement in mental status during the dialysis session, or shortly thereafter.

patient with unknown encephalitis did not improve despite ACV treatment.

Haemodialysis

An average of 3–4 h of haemodialysis decreased CMMG and ACV concentrations by 64 ± 9 and 57 ± 9%, respectively, and symptoms subsided (Figure 4).

Discussion

The main findings of this study are:

(i) In patients developing neuropsychiatric symptoms during ACV therapy, serum concentrations of ACV and of its main metabolite CMMG were increased as compared with those in patients without these symptoms.

Table 3. The most frequent adverse events[a]

	Own study ($n = 37$)	Literature ($n = 28$)	P-value[b]
Age (year)	60 ± 15	53 ± 24	0.50
Sex (Female)	19	14	0.56
[No. of subjects (%)]			
Higher cerebral functions			
Confusion/disorientation	19 (51)	14 (50)	0.56
Visual and/or sound hallucinations	14 (38)	7 (25)	0.30
Restlessness	9 (24)	0 (0)	0.01[c]
Lethargy/drowsy	7 (19)	11 (39)	0.14
Ataxia			
Unsteadiness	8 (22)	1 (4)	0.07
Vertigo	7 (19)	1 (4)	0.10
Involuntary movements			
Myoclonus	6 (16)	5 (18)	0.57
Tremor	4 (11)	6 (21)	0.25
Cranial nerve functions			
Dysarthria/slurred speech	6 (16)	7 (25)	0.34
Disturbances of consciousness			
Somnolence	13 (35)	1 (4)	< 0.01[c]
Unresponsive	6 (16)	3 (11)	0.43
Coma	4 (11)	7 (25)	0.17
Peripheral nerves			
Hyper-reflexia	3 (8)	0 (0)	0.17
GI-intestinal disturbances	14 (38)	7 (25)	0.20
Acute renal failure/increasing creatinine	5 (14)	5 (18)	0.47
Disturbances in lung function	7 (19)	3 (11)	0.34
Fever	5 (15)	1 (4)	0.21
Patients who died in close connection to NPSE	6 (16)	1 (4)	0.14

[a]The commonest neuropsychiatric adverse events were categorized into six main groups reflecting specific functions of the nervous system. Symptoms from other organ systems were also collated. Plus-minus values are means ± SD. NPSE, neuropsychiatric side effects.
[b]The P-values are based on Fishers exact test except for age, which was based on Newman–Keuls test.
[c]Value statistical significant if $P < 0.05$.

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(ii) CMMG was the strongest predictor of neuropsychiatric symptoms.
(iii) Renal insufficiency was present in the majority of patients with neuropsychiatric symptoms.
(iv) The neuropsychiatric symptoms and the CMMG levels decreased simultaneously during haemodialysis.
(v) The symptom profile of the ACV-syndrome is characteristic.

Approximately three publications have been presented per year between 1982 and 2001, in which severe neuropsychiatric events have been associated with ACV therapy. Apparently, neither current treatment recommendations nor available diagnostic tools to discriminate between ACV-associated side effects and herpes virus encephalitis are sufficient to avoid this clinical problem. This is clearly demonstrated by the higher ACV exposure in patients with neuropsychiatric symptoms (Figure 3). However, there were several patients for whom the dose had been adjusted for renal function according to the manufacturer in the Swedish Physicians Desk Reference (the valaciclovir doses are higher in Sweden than in the US). Our data suggest that determination of CMMG levels may be used in the differential diagnosis between neuropsychiatric symptoms due to ACV and other causes with sufficient sensitivity and specificity.

The other predictors of ACV-associated neuropsychiatric symptoms, i.e. ACV exposure, ACV concentration, creatinine clearance and creatinine concentration had sensitivities and specificities lower than CMMG and are clearly not as clinically useful in differential diagnosis. However, the analysis shows that the risk of developing the syndrome is high whenever a patient with poor renal function is exposed to large doses of ACV, either through high doses, i.v. administration or the use of VACV.

An important methodological issue concerns the pharmacokinetics of ACV and CMMG and the data used in this study. First, the samples were collected with a longer interval from the administered dose in patients with symptoms as compared with those without symptoms. Secondly, as almost all patients with symptoms had renal insufficiency the half-lives of both ACV and CMMG were prolonged. We found that the CMMG concentrations were more than seven times higher in patients with CNS symptoms than in those without, a difference that is probably underestimated due to the differences in sampling time. We therefore conclude that the sensitivity and the specificity of CMMG levels for diagnosis of ACV-induced side effects may be even higher. Possibly, the CMMG cut-off value found, 10.8 μmol/l, in the ROC analysis, is slightly overestimated. It is here important to point out that the interpretation of CMMG levels depends on the treatment indication. ACV is vital in herpes encephalitis treatment in which case higher CMMG levels may be acceptable while the limited value of ACV in post-herpetic neuralgia prophylaxis makes even lower CMMG levels unacceptable. In the individual case, the utility of ACV must be evaluated, weighing treatment efficacy and the severity of the condition against the risk of side effects. In this connection it is important to discuss the predicted values of CMMG. The figures given for the positive and negative predicted value depends on the incidence in the population studied which here consists of patients for whom ACV therapy was considered to be so complicated that a concentration was asked for. In the general population the numbers will be much lower.

Another methodological issue is the classification of patients. The retrospective nature of the data reduces the precision with respect to presence of CNS symptoms. However, the evaluation of the symptoms was blinded to avoid bias.

It is most unlikely that the present findings are a random event since the difference in CMMG levels between patients with ACV as the only identified risk factor for CNS symptoms and those without symptoms is so large. A prospective study of the presence of CNS symptoms in relation to CMMG levels may sharpen the difference.

The mechanism by which ACV induces neuropsychiatric side effects is not clear. It is tempting to suggest that CMMG is the cause of this side effect. The evidence for such a causal relation is presently epidemiological. Considerable experimental work remains to be performed to test this hypothesis. An alternative hypothesis is that the other metabolite 8-OH-ACV is the causative agent.

The potential usefulness of the present findings has its basis in the difficulty to differentiate between symptoms of herpes encephalitis and symptoms due to ACV. The reality of the problem is illustrated by earlier case reports and also by cases among our patients, in whom symptoms or worsening of symptoms, apparently associated with ACV, have been misinterpreted as herpes encephalitis and the dosage has been increased, aggravating the clinical condition.

We also found that a single haemodialysis session resulted in a clinical improvement and a reduction in CMMG and ACV concentrations. Thus, haemodialysis can be used both to corroborate the diagnosis of ACV-associated side effects and to treat the condition. This clinical efficiency of haemodialysis has been shown earlier [19,20]. It should be noted that peritoneal dialysis only gave slow recovery and low ACV clearance [10].

In order to provide recommendations for optimal dosing of ACV in renal failure, a systematic investigation of the kinetics of both CMMG and ACV in patients with renal failure is required. The aim is to obtain a maximum production of ACV-tri-phosphate in virus-infected cells at the smallest possible 'cost' in terms of CMMG levels. The present data is not a sufficient basis for such dose recommendations. Obviously, ACV treatment can and should be used to treat herpes encephalitis, even in patients with renal failure, but a monitoring of serum concentrations of CMMG and of CNS symptomathology is warranted. Naturally the doses of ACV should generally be

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reduced in this patient group. It is probably advisable to avoid the use of VACV in patients with renal dysfunction, as unacceptably high concentrations of CMMG are easily attained.

It is concluded that determination of serum concentrations of CMMG may help differentiate between neuropsychiatric side effects of ACV and symptoms of any form of encephalitis. Haemodialysis effectively relieves CNS side effects and decreases serum concentrations of ACV and CMMG. The monitoring of CMMG concentration is suggested in patients with renal dysfunction or when ACV-induced neuropsychiatric side effects are suspected.

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Conflict of interest statement. None declared.

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